

EXPERIMENT 1:

HTML DOCUMENT PARSING

```
import requests
from bs4 import BeautifulSoup

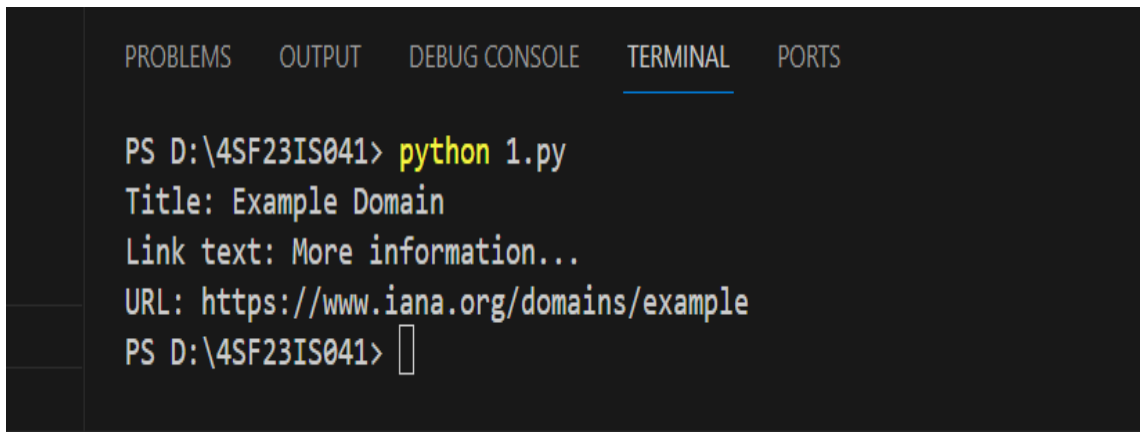
url = "https://www.example.com"
response = requests.get(url)

soup=BeautifulSoup(response.text,"html.parser")

print("Title:",soup.title.string)

for link in soup.find_all('a'):
    print("Link text:",link.text.strip())
    print("URL:",link.get('href'))
```

OUTPUT:

A screenshot of a terminal window with a dark background. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows the following text:

```
PS D:\4SF23IS041> python 1.py
Title: Example Domain
Link text: More information...
URL: https://www.iana.org/domains/example
PS D:\4SF23IS041> 
```

EXPERIMENT 2:

COUNT NUMBER OF WORDS IN A STRING

```
import sys

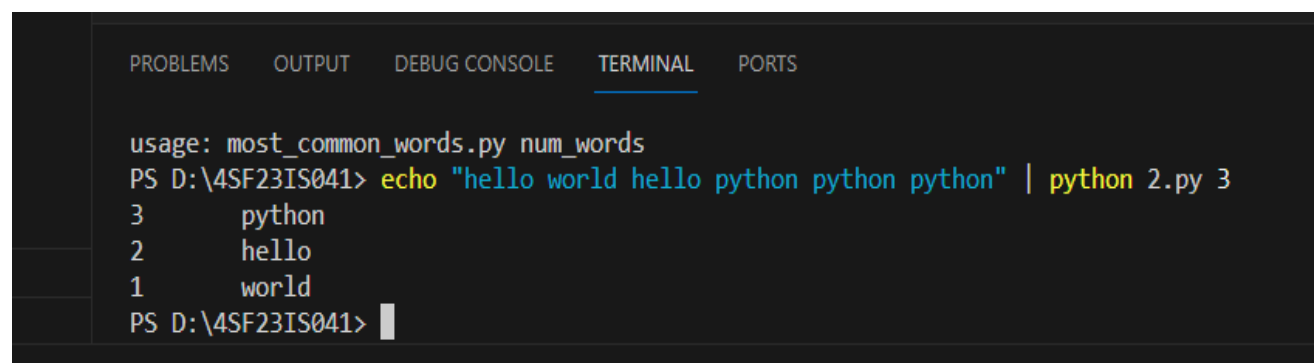
from collections import Counter

try:
    num_words = int(sys.argv[1])
except:
    print("usage: most_common_words.py num_words")
    sys.exit(1)

counter = Counter(word.lower()
    for line in sys.stdin
    for word in line.strip().split()
    if word)

for word, count in counter.most_common(num_words):
    sys.stdout.write(str(count))
    sys.stdout.write("\t")
    sys.stdout.write(word)
    sys.stdout.write("\n")
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

usage: most_common_words.py num_words
PS D:\4SF23IS041> echo "hello world hello python python python" | python 2.py 3
3      python
2      hello
1      world
PS D:\4SF23IS041>
```

EXPERIMENT 3:

NAMED TUPLE

```
from collections import namedtuple
```

```
Student=namedtuple("Student" , ["name","age","marks"])
```

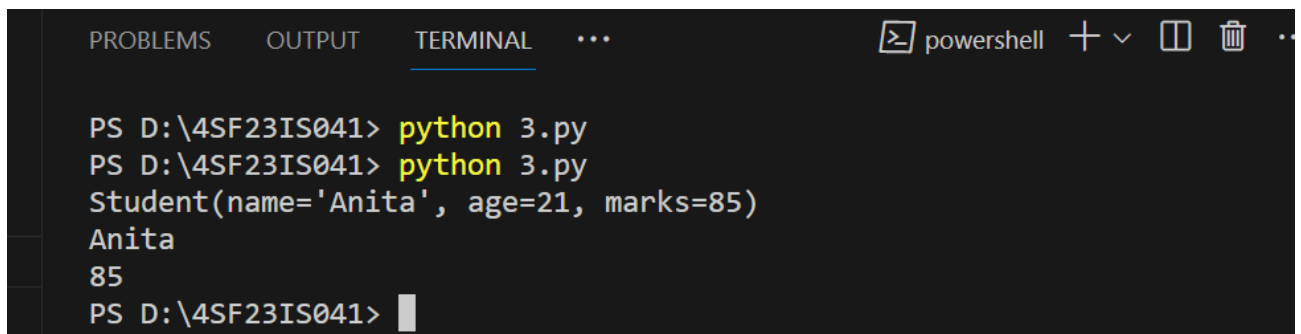
```
s1 = Student("Anita",21,85)
```

```
print(s1)
```

```
print(s1.name)
```

```
print(s1[2])
```

OUTPUT:



```
PROBLEMS OUTPUT TERMINAL ... powershell + v [ ] [X] ..
PS D:\4SF23IS041> python 3.py
PS D:\4SF23IS041> python 3.py
Student(name='Anita', age=21, marks=85)
Anita
85
PS D:\4SF23IS041> 
```

EXPERIMENT 4:

DATACLASS

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class Student:
```

```
    name:str
```

```
    age:int
```

```
    marks: int =0
```

```
s2 = Student("Anita",21,85)
```

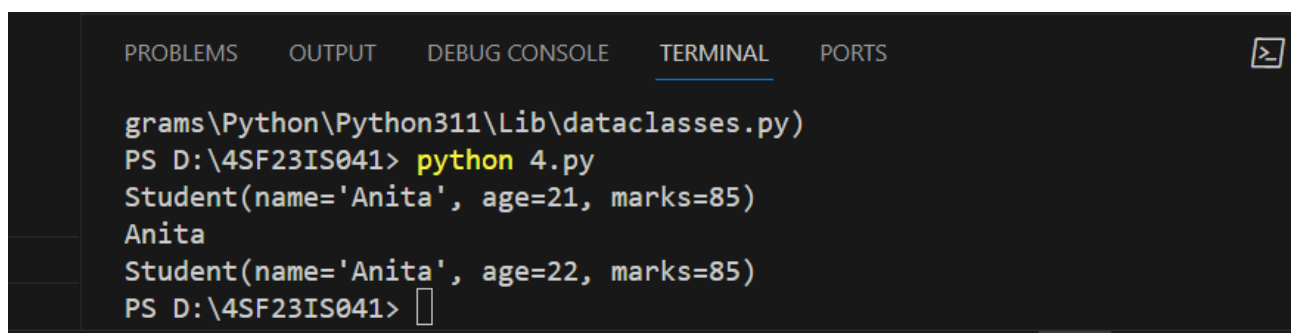
```
print(s2)
```

```
print(s2.name)
```

```
s2.age = 22
```

```
print(s2)
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
```

```
grams\Python\Python311\Lib\dataclasses.py)
PS D:\4SF23IS041> python 4.py
Student(name='Anita', age=21, marks=85)
Anita
Student(name='Anita', age=22, marks=85)
PS D:\4SF23IS041> 
```

EXPERIMENT 5:

COUNT OF EMAIL

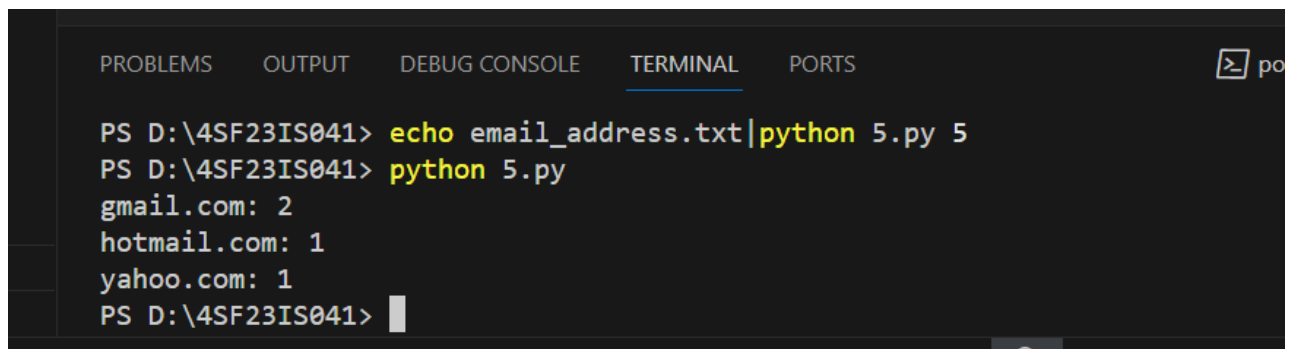
```
from collections import Counter

def get_domain(email_address):
    return email_address.lower().split("@")[-1]

with open('email_address.txt','r') as f:
    domain_counts = Counter(get_domain(line.strip())
                             for line in f
                             if "@" in line)

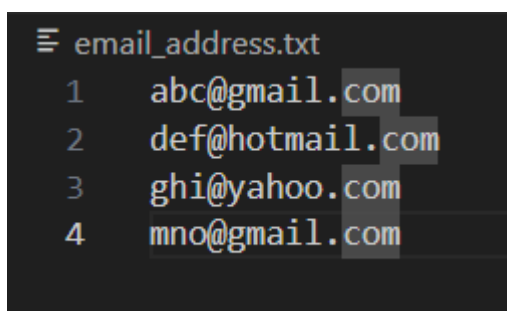
for domain, count in domain_counts.items():
    print(f"{domain}: {count}")
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\4SF23IS041> echo email_address.txt | python 5.py 5
PS D:\4SF23IS041> python 5.py
gmail.com: 2
hotmail.com: 1
yahoo.com: 1
PS D:\4SF23IS041> 
```



```
email_address.txt
1 abc@gmail.com
2 def@hotmail.com
3 ghi@yahoo.com
4 mno@gmail.com
```